

get chiplets from different brands that specialise in specific functions, and package them together—provided they adhere to some basic standards. It is like putting together a team of experts.

- **Better optimisation.** Each chiplet is typically optimised to do its function in the best possible way, with lower power consumption and better performance. If integrated properly, the final device is also super optimised.

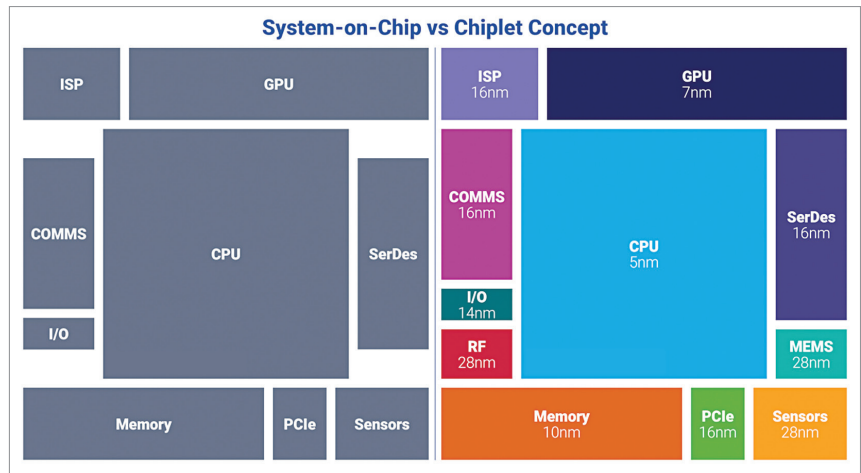
- **Flexibility and scalability.** Since chiplets are modular components, they can be easily added or removed to adjust the functionality and performance of the SoC. You can also replace or customise a chiplet, without affecting the rest of the SoC. This speeds up innovation.

How are they combined?

- Chiplets are assembled into a package and interconnected using a die-to-die interconnect framework to make an SoC.

- **Interconnects.** Interconnects enable communication between the chiplets. High-speed, low-latency and power-efficient standards like Universal Chiplet Interconnect Express (UCIe), Bunch of Wires (BoW), Open High Bandwidth Interconnect (OpenHBI), and Optical Internet Forum Extra Short Reach (OIF XSR) play a major role in the success of chiplet-based designs.

- **Packaging.** Varied methods are used to assemble chiplets into a larger IC, often known as a multi-chip module, a hybrid IC or an advanced package. In 2.5D packaging, the chiplets are placed side-by-side on a silicon interposer to achieve high interconnect density. In 3D packaging, the chiplets are stacked vertically to achieve the shortest interconnect paths and smallest package footprint. Other packaging technologies include system-in-package (SiP), flip chip



Traditional SoC vs chiplets (Courtesy: IDTechX)

chip scale package (FCCSP), flip chip ball grid array (FCBGA), wafer-level chip scale package (WLCSP), and fan-out (FO).

- **Electronic design automation (EDA) tools.** Modern EDA tools/workflows from companies like Synopsis and Cadence allow engineers to easily integrate modular and heterogeneous chiplets into a package, optimise the SoC design, and verify whether the chiplets work together seamlessly and meet the required performance and reliability standards.

What are the current challenges?

- Need for more industry-wide interoperability standards
- Thermal management issues arising from the proximity of chiplets in a package
- Latency and power consumption issues
- Design and testing issues

Spells opportunity for...

- EDA tool providers
- Chiplet designers
- Manufacturers and fabricators
- Chiplet packaging and testing service providers
- OEMs and system integrators

arising from the integration of chiplets from multiple vendors

- Intellectual property protection and licensing complexities.

How is the chiplets market?

- There is a surge in demand for high-performance computing (HPC) and data centres, primarily driven by the rise of artificial intelligence (AI), military and aerospace applications, industrial automation, smart homes, and other Internet of Things (IoT) applications.

- Off-the-shelf chiplets are in focus, enabling companies to assemble flexible and scalable HPC systems swiftly. Advanced packaging technologies and EDA tools are further fuelling this growth.

- We see companies ranging from tech giants like Apple, IBM, AMD, Intel, and Nvidia, to their challengers like Canadian company Tenstorrent, and startups like Kerala-based Netrasemi using chiplet technologies to develop high-performance chips.

- According to a report by Fortune Business Insights, the chiplets market is projected to grow from US\$44.82 billion in 2024 to US\$233.81 billion by 2032, exhibiting a CAGR of 22.9% during the forecast period. **EFY**